

Session 19:

AI Careers & The Future of Work

you've explored how AI reads, sees, hears, combines senses, and transforms entire industries. You've learned that AI is a pattern-matching engine - powerful, but not alive.

Today, we'll discover how AI is reshaping work, which human skills will always be in demand, and what careers are emerging for those who understand AI from the inside. Then, you'll step into the role of a builder and create your own personalized web app with Lovable



Objectives

By the end of this session, you will be able to:

01

Explain AI & Work

Understand how AI is changing work.

02

Identify Skills

Essential human skills for thriving in the AI era.

03

Explore AI Careers

Describe a variety of emerging AI careers across four exciting lanes.

04

Build a Real App

Create a simple, personalized web app using Lovable—no code required.

05

Human-AI Collaboration

Explain why human-AI collaboration is the winning formula for the future.

Recap

Last session (S18): You discovered AI in healthcare, finance, education, entertainment, and transportation and identified a real-life problem AI could help solve. **Today, you'll build the solution.**

Domain	Framework	Tool You Used
NLP	Break → Understand → Generate	QuillBot, NotebookLM
Computer Vision	See → Process → Act	Teachable Machine
Voice AI	Hear → Process → Speak	ElevenLabs
Multimodal AI	Sense → Think → Act	Gemini, Veo 3, Pictory AI
Across Industries	Sense → Think → Act applied	Gemini + NotebookLM

Warm-Up: The Big Question

Think about a job you admire. **Teacher? Doctor? Game designer?**
YouTuber? Chef? Astronaut?

Now imagine the year is **2040**. Will that job still exist? Will a robot do it for you? Or will AI be your assistant, making the job more exciting?

i 🤖 **Here's a secret:** Whatever job you dream of, there's almost certainly a way AI will help you do it better. Today we'll explore how.





How AI Is Changing Work

Automates

Handles repetitive, time-consuming tasks so humans don't have to.

Augments

Enhances human skills to make us smarter, faster, and more creative.

Creates

Entirely new careers that never existed before.

AI is not just doing our old jobs, it's reshaping how work happens.

Every technology shift creates new jobs and changes old ones. A hundred years ago, most people worked on farms. Fifty years ago, factories employed millions. Today, AI is the biggest change agent and the good news is that you are perfectly positioned because you understand AI from the inside.

① The future of work is not *human versus machine*; it's **human with machine**.



Real-World Example: The Cashier

Old Way

One cashier scans every item.
Repetitive. Tiring. Limited to one customer at a time.

- Scanning every item manually
- Counting change
- Processing payments

New Way

Self-checkout kiosks (AI) scan items. One human oversees six kiosks, solves problems, and provides friendly help.

- **AI handles:** Scanning, counting, payment math
- **Human handles:** Problem-solving, empathy, complex questions



Key Insight: AI removes the boring parts. Humans focus on what we do best - connecting, caring, and solving complex problems.

Skills You Need in the AI Era

AI is great at patterns. You are great at being human. These skills are your superpowers.

Thinking Skills

- **Critical Thinking** - Question what AI tells you.
- **Problem-Solving** - Break big challenges into steps.
- **Judgment** - Decide what's right, fair, and wise.
- **Creativity** - Imagine what has never existed.

Human Skills

- **Communication** - Explain ideas clearly to AI and people.
- **Empathy** - Understand how others feel.
- **Ethics** - Know the difference between *can we?* and *should we?*
- **AI Literacy** - Use AI tools effectively and responsibly.

 These skills work together. The best AI users are both sharp thinkers and kind humans.

Mini Activity 1: Spot the Skill

Read each scenario. Which of the eight skills is being used the most?

#	Scenario	Primary Skill
1	A student learns a brand new AI video tool in one afternoon to finish a school project.	
2	A game designer imagines a character that has never been seen before - a clockwork dragon that breathes steam.	
3	You ask AI for advice, but then think: "Is this actually a good idea for my situation?"	
4	You write a very detailed prompt so AI understands exactly the tone you want.	
5	You realize an AI tool might be unfair to some users and decide not to use it for that purpose.	

Mini Activity 1: Answer Key

Read each scenario. Which of the eight skills is being used the most?

#	Primary Skill
1	AI Literacy
2	Creativity
3	Judgment
4	Communication
5	Ethics

Emerging AI Careers: The Four Lanes

AI jobs fall into four exciting lanes. Today, you'll meet a few examples from each but there are **hundreds more!**



The Builders

Build the AI tools, models, and robots.

Prompt Engineer, ML Engineer, AI Engineer, Robotics Engineer...



The Deciders

Figure out what problems AI should solve and how.

Data Scientist, AI Product Manager, AI Solutions Architect, CAIO...



The Guardians

Ensure AI is fair, safe, legal, and used for good.

AI Ethicist, AI Compliance Analyst, AI Policy Advisor...



The Empowered

Use AI to supercharge their existing job or passion.

AI-Powered Teacher, AI-Assisted Artist, AI-Augmented Doctor...

 **Remember:** These are just examples. AI is creating new roles in sports, farming, fashion, space exploration, and everywhere else you can imagine.



The Builder : Prompt Engineer

Writes precise, creative instructions to get the absolute best results from generative AI. They know the "magic words" that turn a vague idea into a stunning image or a helpful response.

What They Do

Craft detailed, structured prompts that guide AI models to produce exactly the right output—images, text, code, or audio.

Why It Matters

The quality of AI output depends almost entirely on the quality of the prompt. A great prompt engineer is worth their weight in gold.

Analogy

A word wizard who speaks fluent AI.

The Builder : Machine Learning Engineer

Designs and trains the models that learn patterns from data. They build the brains behind recommendation systems, fraud detectors, and voice assistants.

What They Do

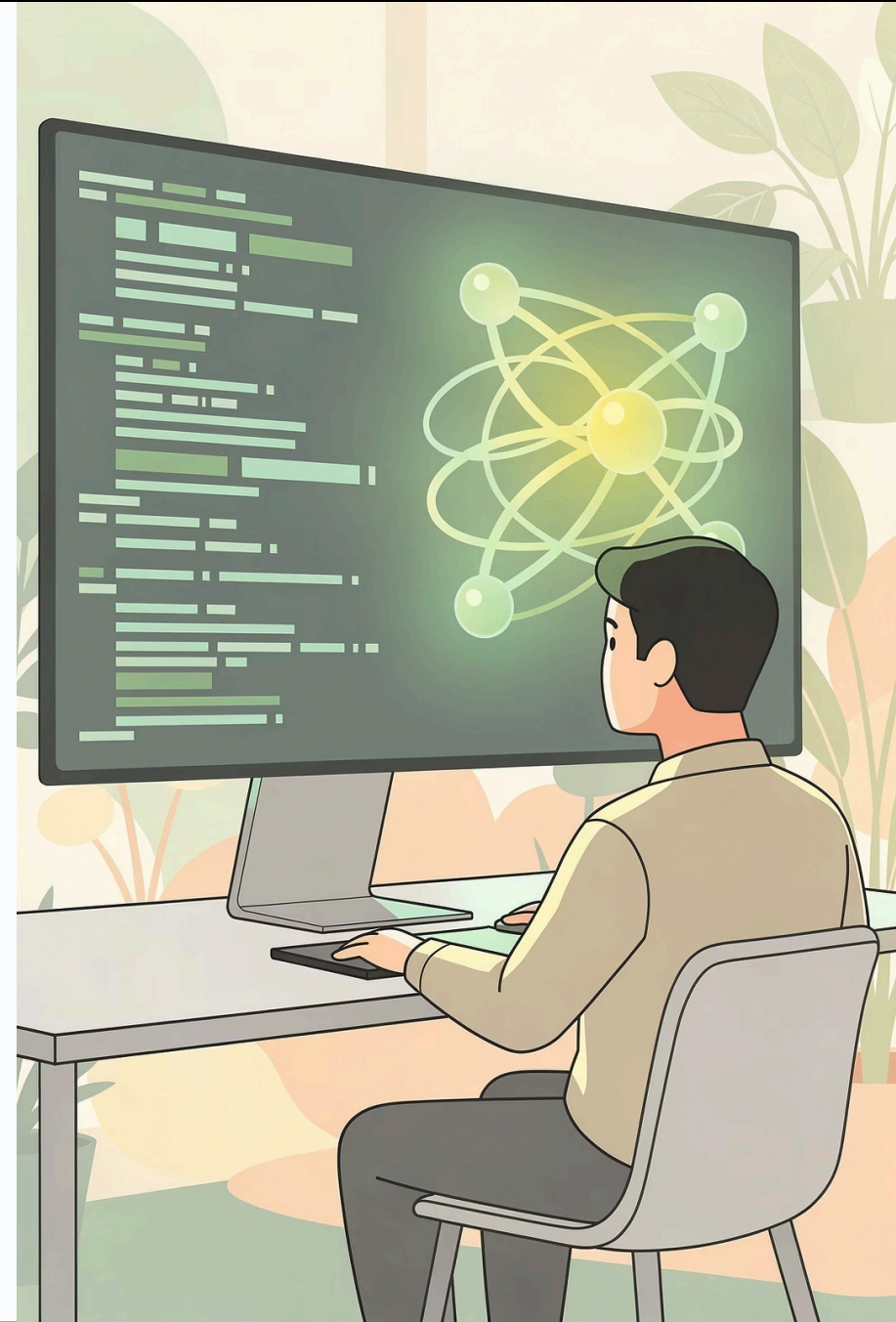
Write algorithms that allow computers to learn from experience getting better over time without being explicitly programmed for every scenario.

Where You See Their Work

Netflix recommendations, spam filters, face recognition, and the AI that detects credit card fraud.

Analogy

A teacher who trains AI using millions of flashcards.





The Builder : AI Engineer

Takes trained AI models and builds them into real, working software applications. They make sure the AI works reliably inside an app, a website, or a smart device.

What They Do

AI Engineers sit between the research world (where models are created) and the real world (where people use apps). They make the magic accessible.

Where You See Their Work

- Software development
- API integration
- Cloud deployment
- Testing & reliability

Analogy

A contractor who turns blueprints (models) into a finished house (app).

The Builder : Robotics Engineer

Combines AI with physical machines that can sense and move in the real world. They build self-driving cars, warehouse robots, surgical assistants, and Mars rovers.



Autonomous Vehicles

Teaching cars to navigate roads, detect obstacles, and make split-second decisions safely.



Industrial Robots

Building warehouse robots that pick, pack, and sort thousands of items per hour.



Space Exploration

Designing Mars rovers that navigate alien terrain autonomously, millions of miles from Earth.



The Decider : Data Scientist

Analyzes huge amounts of messy data to find hidden patterns and stories. Their insights help companies decide what to build, what to sell, and what customers really want.

The Data Detective

Sifts through millions of data points to find the signal in the noise - the insight that changes a business decision.

Tools of the Trade

Python, SQL, machine learning libraries, data visualization tools, and statistical analysis.

Analogy

A pattern hunter who turns numbers into smart decisions.





The Decider : AI Product Manager

Decides what AI product to build and why people will want to use it. They understand both the technology and the user's needs, acting as a bridge between engineers and the real world.

What They Balance

User Needs

What problem does this solve for real people?

Technical Feasibility

Can the engineering team actually build this?

Business Value

Will this create value and be sustainable?

Analogy

A ship captain who chooses the destination and keeps everyone on course.

The Decider : AI Solutions Architect

Designs the big-picture blueprint for how different AI systems will work together inside a company. They ensure everything connects smoothly, securely, and efficiently.

Understand the Problem

Meet with business leaders to understand what challenges need AI solutions.

Design the Blueprint

Map out how data flows, which AI models are needed, and how systems connect.

Guide Implementation

Work with engineers to ensure the architecture is built correctly and securely.

Optimize & Scale

Continuously improve the system as the company grows and needs evolve.

Analogy: A city planner who designs where the roads, power lines, and buildings should go.



The Decider : Chief AI Officer (CAIO)

The top executive responsible for an entire company's AI strategy. They decide where to invest, how to use AI responsibly, and how to stay ahead of competitors.

Strategic Vision

Sets the long-term direction for how AI will transform the entire organization.

Investment Decisions

Decides which AI projects get funded and which tools the company adopts.

Responsible Leadership

Ensures AI is used ethically, fairly, and in compliance with regulations.

Competitive Edge

Keeps the company ahead by spotting AI trends before competitors do.

Analogy: The general manager of a sports team - setting the vision and making the biggest decisions.





The Guardian : AI Ethicist

Ensures AI is fair, unbiased, and doesn't cause harm. They test for hidden discrimination and create guidelines for responsible AI use.

The Questions They Ask

- Does this AI treat all groups of people equally?
- Could this system be used to harm someone?
- Is the AI transparent about how it makes decisions?
- Who is accountable when AI makes a mistake?

Why This Role Matters

As AI becomes more powerful, the potential for harm grows too. AI Ethicists are the conscience of the tech industry - ensuring progress doesn't come at the cost of fairness.

  **Analogy:** A fairness detective who investigates whether AI is treating everyone equally.

The Guardian : AI Compliance Analyst

Ensures that AI systems follow all the laws, rules, and company policies. They check that an AI tool doesn't violate privacy rights or break new government regulations.

Privacy Protection

Verifies that AI systems handle personal data according to laws like GDPR and CCPA.

Regulatory Compliance

Tracks new AI laws and ensures company products meet all government requirements.

Internal Policy

Creates and enforces company-wide rules for how AI tools can and cannot be used.

Risk Assessment

Identifies potential legal or reputational risks before an AI product launches.

Analogy: A safety inspector who makes sure a building meets all the legal codes before people move in.

The Guardian : AI Policy Advisor

Works with governments, non-profits, or large companies to help write the "rules of the road" for AI. They research how AI affects society and recommend laws that keep people safe while encouraging innovation.



Research

Studies how AI impacts employment, privacy, democracy, and social equality.



Advise

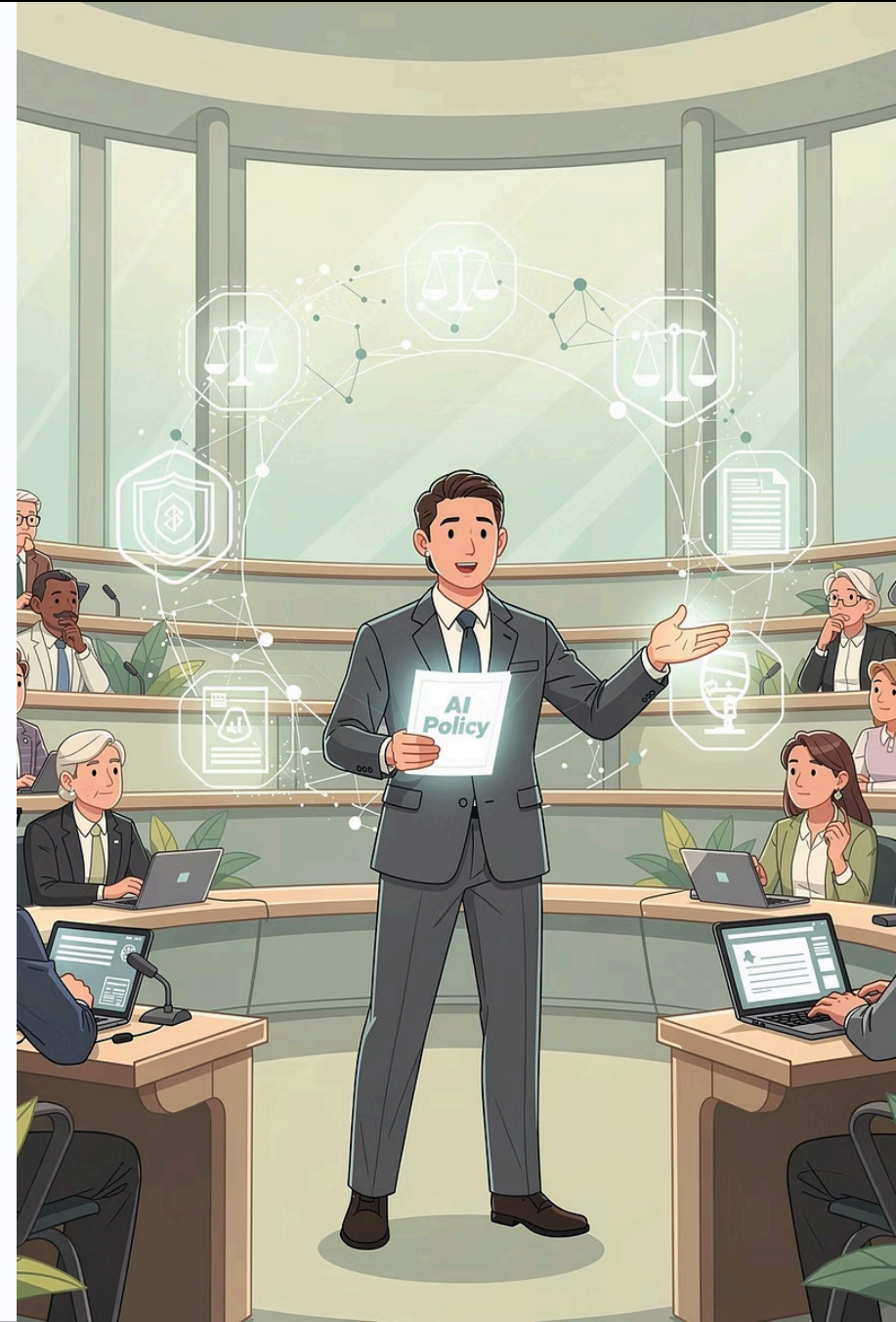
Briefs lawmakers and executives on what AI can do and what guardrails it needs.



Shape Policy

Helps draft regulations that balance innovation with protection of citizens' rights.

Analogy: A lawmaker's expert guide who explains what AI can do and what limits it needs.



The Empowered: AI Enthusiasts

You don't need to be an engineer to use AI. **Every career can be augmented by AI.** These are the artists, doctors, teachers, chefs, farmers, athletes, etc who use AI as a creative partner to do their jobs better, faster, and with more imagination.

Role (Example)	How They Use AI
AI-Powered Teacher	Uses AI to generate personalized practice problems for every student in the class.
AI-Assisted Artist	Uses AI to brainstorm color palettes, sketch variations, or create entire music videos.
AI-Augmented Doctor	Uses AI to highlight suspicious areas in a medical scan that the human eye might miss.
YOU (AI-Curious Student)	Uses AI to explain tricky homework, plan a project, or build an app - like you will today!

The Winning Formula: AI + Human Judgment

Let's see how human-AI collaboration works in a real scenario:

Designing a New Playground.

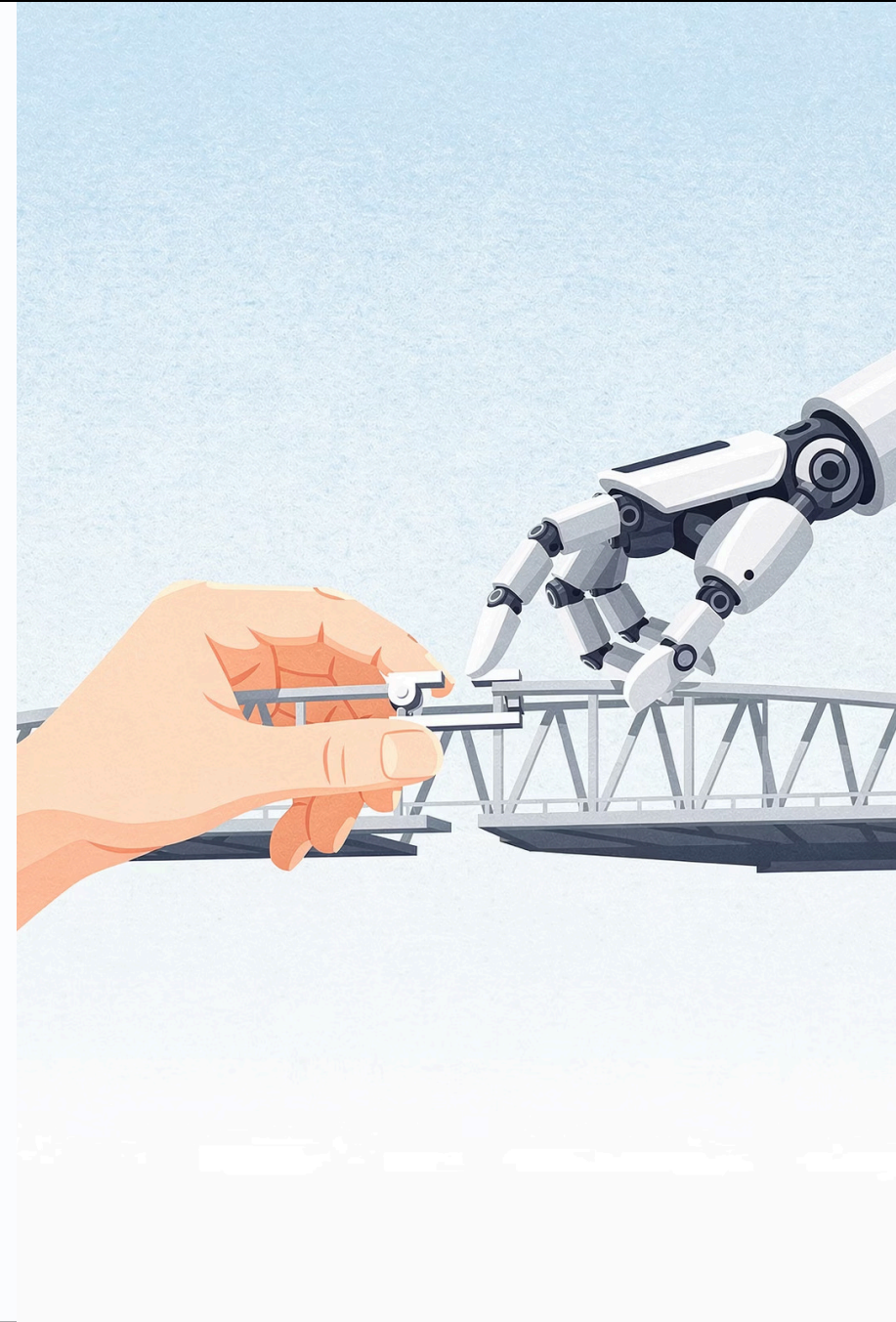
AI (Fast & Tireless)

- Generates 20 playground layouts in seconds.
- Checks safety regulations and compliance rules.
- Estimates material costs for each design.

Human (Wise & Empathetic)

- Knows that kids in this neighborhood love climbing structures.
- Chooses the design with the most shade for hot days.
- Adds a unique mural designed by local art students.

 **The Result:** A playground that is efficient, safe, and full of heart - something neither could create alone. **Human-in-the-Loop = The Ultimate Career Skill.**



Mini Activity 2: The AI Ethics Committee

Scenario: A company has built an AI tool that scans student essays and gives instant feedback on grammar, clarity, and argument strength. They want to launch it in schools.

① **Your Role:** You are the AI Ethics Committee. Your job is to protect students. For each decision, ask: *Does this help students learn, or does it harm them? Where must a human stay in the loop?*

Review each proposed AI feature below and cast your vote -  **Approve** or  **Reject**. Be ready to defend your reasoning.

Mini Activity 2: The AI Ethics Committee

1

Auto-Grade Essays

The AI will assign a grade (A, B, C, D, F) to every essay automatically.

Your Vote: ☒ Approve or ☐ Reject?

2

Highlight & Suggest

The AI will highlight weak sentences and suggest improvements.

Your Vote: ☒ Approve or ☐ Reject?

3

Weekly Parent Rankings

The AI will send a weekly report to parents ranking their child against classmates.

Your Vote: ☒ Approve or ☐ Reject?

4

Flag AI-Written Essays

The AI will flag essays that seem written entirely by AI for teacher review.

Your Vote: ☒ Approve or ☐ Reject?

5

Human Review Required

A human teacher must review every essay before a final grade is given.

Your Vote: ☒ Approve or ☐ Reject?

Mini Activity 2 : Answer Key

1

 Reject

Judgment required. A grade affects a student's future. Only a teacher who knows the student's effort, growth, and context can assign a fair grade. AI lacks empathy and understanding of individual circumstances.

2

 Approve

Augmentation, not replacement. This helps students learn and revise. The student still decides what to change. It's like a smart spell-checker - a tool in service of the learner.

3

 Reject

Ethics and empathy violation. Ranking students publicly causes anxiety, shame, and unhealthy competition. It reduces learning to a number. Human teachers know how to communicate progress with care.

4

 Approve

Human-in-the-Loop. This is a helpful flag, but a human must investigate. AI may falsely accuse a student with a unique writing style. The teacher makes the final call - always.

5

 Approve

Non-negotiable. This is the ultimate Human-in-the-Loop safeguard. AI assists; humans decide what matters. No grade should ever be issued without a teacher's informed judgment.

 This activity simulates the judgment calls made by AI Ethicists and Product Managers. Knowing where AI assists and where humans decide is a core skill for the AI era.

Stay Curious!

The Adaptability Mindset

The Careers of 2040

Many of the careers you'll hold might not exist yet. The tools you'll use haven't been invented. That's exciting, not scary.

Always in Demand

Curiosity and adaptability will always be valued - no matter how much technology changes around us.

 "The illiterate of the 21st century will not be those who cannot read and write, but those who cannot learn, unlearn, and relearn."

- Alvin Toffler





Hands-On Activity: Build Your Own App with Lovable

Goal: Turn a real-life problem into a working, personalized tool using AI.

Tool: Lovable (lovable.dev) - A no-code AI app builder that turns your description into a real, working web application.

1

Pick an Idea

Choose one idea from four different life categories.

2

Describe It

Tell Lovable your idea in one clear sentence.

3

Watch AI Build

See your tool come to life instantly.

4

Improve It

Make one improvement using a follow-up prompt.

Hands-On Activity: Choose Your Idea

Pick one idea from below or choose your own problem you want to solve.

School

Problem: I forget what homework is due each day.

"A simple weekly planner with a text box for each weekday where I can type my assignments."

Home

Problem: My family argues about whose turn it is to do chores.

"A clean chart showing chores for three people: Alex, Jamie, and Taylor, with empty checkboxes next to each task."

Hobby

Problem: I lose track of what I practiced on guitar.

"A practice log with a text box for each day where I can write what I worked on."

Well-being

Problem: I want to remember one good thing each day.

"A simple page with a large text box labeled 'One good thing today' and a button that says 'Save.'"

Step-by-Step: Build Your Tool

Open Lovable

Go to **lovable.dev** and sign in. Click "**New Project.**"

Describe Your App

Type your chosen one-sentence prompt into Lovable.

AI Builds. You Explore.

Watch Lovable generate your tool. Type in the boxes. Check the boxes.

Make Improvement

Ask Lovable to make a change using one of the example prompts.

Example Improvement Prompts

Add Something

"Add a section at the bottom called 'Notes.'"

Change the Look

"Change the background to a soft blue gradient."

Add a Message

"Add a small line at the top: 'You've got this!'"

Summary

AI & Work

AI automates repetitive tasks; humans provide creativity, judgment, and empathy.

Eight Essential Skills

Thinking Skills + Human Skills = Your AI-era superpowers.

Four Career Lanes

Makers, Deciders, Guardians, Empowered—and hundreds more roles!

Collaboration

AI + Human Judgment = The winning formula for the future.

Limitless Possibilities

The roles we met are just the tip of the iceberg.

You Built It!

You used Lovable to create a personalized app from your own idea.

 **Most Important Idea:** You are not competing with AI. You are learning to **lead with AI**. The future needs your uniquely human skills.



Homework: Discover Your Lane

Today you explored four career lanes. Now reflect on where **you** might fit. Complete all three parts before next class.



Builders

Build the AI tools, models, and robots.



Deciders

Figure out what problems AI should solve.



Guardians

Ensure AI is fair, safe, and used for good.



Empowered

Use AI to supercharge an existing career.



Choose the lane that sounds most interesting to you right now, then write **one sentence** explaining why.

Part 2: Connect a Skill

Look back at the **skills** from today. Pick **one skill** that is most important for your chosen lane. Write it down and explain your choice in one sentence.

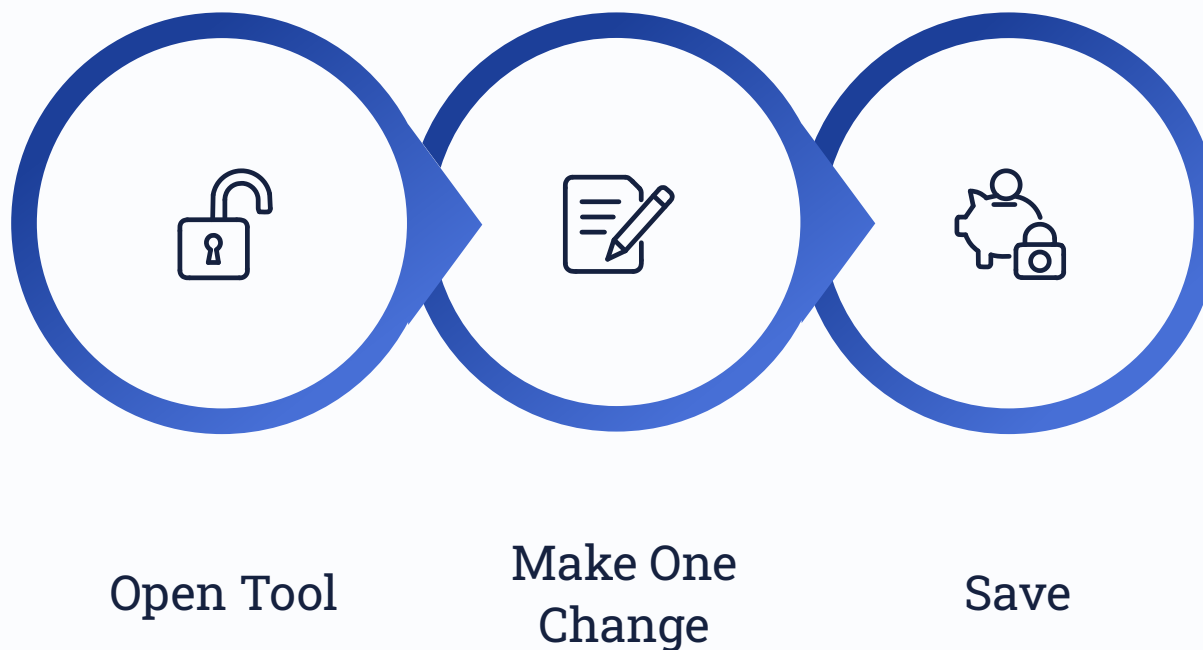
Skill You Picked	Why It Matters for This Lane



Think carefully - there's no wrong answer. The best skill is the one that genuinely connects to the lane **you** chose and the work you'd actually want to do.

Part 3: Refine Your Tool

Open your **Lovable** tool one more time. Make **one small improvement** - change a color, add a word, or adjust the layout. Every iteration makes you a better AI collaborator.



Next Session Preview

Session 20 : AI Bias & Fairness

What You'll Explore

- Why do AI systems sometimes treat people unfairly?
- Where does bias come from?
- How can we build fairer AI for everyone?

